

Project Team Meeting: Agenda

Cross-Generator Generalization in Deepfake Detection

IE7374: Generative AI, Northeastern University

Date: _____	Time: approximately 45 minutes
Facilitator: Dominic Rivas	Prepared by: Dominic Rivas
Attendees: Dominic Rivas, Jonathan Regalado, Lyxelis Navarro, Obinna Okonkwo, Sagar Ayare	
Objective: Confirm the model and dataset scope, divide the eight training runs, and confirm member roles.	

Agenda

No.	Agenda item	Lead	Time	Desired outcome
1	Project summary and goal	Rivas	5 min	Shared understanding
2	Review of research findings (pre-read Sections 1 to 5)	Rivas	10 min	Context agreed
3	Open decisions (Section 6)	All	10 min	Decisions recorded
4	Split the eight training runs and confirm roles (Section 7)	All	10 min	Owners assigned
5	Risks and mitigations (Section 8)	All	10 min	Risks logged

Background Pre-Read

This pre-read summarizes the research behind the project so the meeting can focus on decisions. All figures are cited and listed in the References at the end.

Key terms

AUC	Area Under the ROC Curve (ROC stands for Receiver Operating Characteristic). A single score from 0 to 100 for how well the model separates real from fake across every decision threshold: 100 is perfect and 50 is no better than a coin flip. It is the standard metric in this field because it does not depend on choosing one cutoff. AUC and AUROC mean the same thing.
CNN	Convolutional Neural Network, the standard image-classification model we use.
GAN	Generative Adversarial Network, a common method for generating deepfakes.
FF++	FaceForensics++, our primary dataset of real and manipulated face videos.
DFDC	DeepFake Detection Challenge, a second dataset used only as an optional test.
LOMO	Leave-One-Manipulation-Out, our protocol: train on 3 of the 4 methods, test on the 4th.
In-distribution	Tested on a method the model was trained on (the easier case).
Cross-generator	Tested on a method the model was not trained on (the harder case).

1. Why this matters

Fewer than 1% of people can reliably identify a deepfake [1]. Deepfakes are already used for fraud, harassment, and misinformation, and the U.S. Government Accountability Office and UNESCO identify them as a threat to information integrity [2], [3]. Detection lags generation, because each new generation method produces a new pattern of artifacts that existing detectors have not seen. A clear recent example is the shift from GAN-based methods to diffusion-based methods [6], [13]. This remains a Generative-AI project even though we do not train a generator: the content we detect is the output of generative models, and the central challenge is the distribution shift from one generation method to another.

2. Key finding from the literature

Detectors that are strong on a method they were trained on lose much of their accuracy on methods they were not trained on. Several independent studies report the same pattern [4], [5], [6], [7]:

Setting	In-distribution	Unseen method or dataset	Source
FaceForensics++ to Celeb-DF	99.7 AUC	48.2 AUC (below random)	Celeb-DF, 2020 [4]
FaceForensics++ leave-one-out (Xception)	about 96 AUC	77.9 average; 51.2 on FaceSwap	LipForensics, 2021 [5]
GAN-trained model on diffusion images	97.4 AUC	78.6 AUC (detects 26% of fakes)	Ricker et al., 2024 [6]
FaceForensics++ to DFDC	0.96 AUC	about 0.71 AUC	DeepfakeBench, 2023 [7]

The clearest single result is the 48.2 AUC for an FaceForensics++ detector tested on Celeb-DF: a value below 50 means the detector performs worse than random guessing on data from a new source [4].

3. Dataset: FaceForensics++

FaceForensics++ contains 1,000 real videos and 4,000 manipulated videos, for 5,000 in total. The same 1,000 originals are altered by four different methods, which form a clean two-by-two grouping that is the basis of our transfer hypotheses [8]:

	Face-swap (changes identity)	Reenactment (keeps identity)
Learning-based	DeepFakes	NeuralTextures
Graphics-based	FaceSwap	Face2Face

- We use the c23 (high-quality) version, about 20 GB in total, which fits our storage and preprocessing budget. The raw version is roughly 1 TB and is not needed [8].
- The official 720 / 140 / 140 split is identity-disjoint by design: the same person never appears in both training and test data. This avoids a common cause of inflated results, where the model recognizes the person rather than the manipulation [8].
- The data is fake-heavy (1 real to 4 fake), so we balance it per method, 1,000 real against 1,000 fake [8].

FaceForensics++ is preferable to a single large dataset because it provides four cleanly separated manipulation methods built on the same faces. This lets us separate the effect of an unseen manipulation method from the effect of an unseen person, which is exactly what the project measures [8].

4. Method

We train two CNN models, EfficientNet and XceptionNet, both pretrained on ImageNet and fine-tuned rather than trained from scratch. XceptionNet is the standard FaceForensics++ baseline [8], [9], and EfficientNet is the backbone used by the DeepFake Detection Challenge winner [10], [11]. The protocol is Leave-One-Manipulation-Out: we train on three of the four methods and test on the fourth, rotating which method is held out. With two models this gives four setups times two models, or eight training runs [5], [7]. The main deliverable is a transfer matrix, comparing the method trained on against the method tested on for every pair, with the in-distribution versus cross-generator gap as the headline result.

The work is feasible on a single 12 GB GPU. After cropping faces once to about 80,000 to 160,000 cached images, the eight runs take days rather than weeks; the bottleneck is preprocessing, not GPU time [7]. Because the eight runs are independent, members can each own one or two, which makes the workload easy to divide evenly.

5. Expected results

- In-distribution accuracy of about 0.96 AUC, the routine case [7].
- Held-out (unseen) accuracy of about 0.55 to 0.85 AUC, a drop of roughly 15 to 40 points. This gap is the result; a large gap is a meaningful finding, not a failure [5], [7].
- NeuralTextures is expected to be the hardest method to detect when held out, because it is the subtlest manipulation [5].
- FaceSwap is expected to be the weakest training source, because its graphics-based artifacts transfer poorly to learning-based methods [5].
- DeepFakes and Face2Face are expected to transfer best to each other, because their artifacts are most similar [5].

One caveat keeps the comparison fair: most published numbers train on one method, whereas we train on three. The added variety should make our gap somewhat smaller than the published single-method figures, and we will state this in the report [5].

6. Decisions to settle today

No.	Decision	Recommendation
1	Run both EfficientNet and XceptionNet, or one?	Both, to allow an architecture comparison
2	Pretrained or trained from scratch?	Pretrained (faster; course-recommended)
3	Is the DFDC cross-dataset test in scope?	Stretch goal only (preprocessing time)
4	Who owns which of the eight training runs?	Split two models by four folds across the team
5	Assign roles (Section 7)	One area is still open
6	Official FF++ request form or Kaggle mirror?	Form for a clean license; mirror is faster

Decisions 1 to 3 were pre-selected for the draft proposal; this meeting is the opportunity to confirm or change them.

7. Roles

To be assigned during the meeting. Each member co-writes the report section tied to their area and reviews one other section.

Member	Area	Responsibilities
Dominic Rivas		
Jonathan Regalado		
Lyxelis Navarro		
Obinna Okonkwo		
Sagar Ayare		

8. Risks and mitigations

- Identity leakage, where the same face appears in training and test, inflates results. Mitigation: the FaceForensics++ official split already prevents this, and the splits are audited to confirm [8].
- Re-decoding video every training epoch wastes time. Mitigation: crop faces once, cache them to disk, and have everyone train from the cache.
- Using the held-out method during tuning would invalidate the result. Mitigation: keep the held-out method's files separate until final evaluation.
- Comparing our Leave-One-Manipulation-Out numbers to single-method papers is not a like-for-like comparison. Mitigation: state the caveat from Section 5 [5].
- The full DFDC set is about 470 GB [14]. Mitigation: use the Preview subset of about 5,000 clips only, and only if time allows [12].

References

- [1] Zupan, C. (2025). (Deep)fake news: Recent data reveals gaps between perception and reality. Mimecast. <https://www.mimecast.com/blog/deepfake-news-recent-data-reveals-gaps-between-perception-and-reality/>
- [2] U.S. Government Accountability Office. (2024). Science and tech spotlight: Combating deepfakes (GAO-24-107292). <https://www.gao.gov/products/gao-24-107292>
- [3] Naffi, N. (2025). Deepfakes and the crisis of knowing. UNESCO. <https://www.unesco.org/en/articles/deepfakes-and-crisis-knowing>
- [4] Li, Y., Yang, X., Sun, P., Qi, H., and Lyu, S. (2020). Celeb-DF: A large-scale challenging dataset for DeepFake forensics. CVPR. <https://arxiv.org/abs/1909.12962>
- [5] Haliassos, A., Vougioukas, K., Petridis, S., and Pantic, M. (2021). Lips don't lie: A generalisable and robust approach to face forgery detection. CVPR. <https://arxiv.org/abs/2012.07657>
- [6] Ricker, J., Damm, S., Holz, T., and Fischer, A. (2024). Towards the detection of diffusion model deepfakes. VISAPP. <https://arxiv.org/abs/2210.14571>
- [7] Yan, Z., Zhang, Y., Yuan, X., Lyu, S., and Wu, B. (2023). DeepfakeBench: A comprehensive benchmark of deepfake detection. NeurIPS Datasets and Benchmarks Track. <https://arxiv.org/abs/2307.01426>
- [8] Rossler, A., Cozzolino, D., Verdoliva, L., Riess, C., Thies, J., and Niessner, M. (2019). FaceForensics++: Learning to detect manipulated facial images. ICCV. <https://arxiv.org/abs/1901.08971>
- [9] Chollet, F. (2017). Xception: Deep learning with depthwise separable convolutions. CVPR. <https://arxiv.org/abs/1610.02357>

- [10] Tan, M., and Le, Q. V. (2019). EfficientNet: Rethinking model scaling for convolutional neural networks. ICML. <https://arxiv.org/abs/1905.11946>
- [11] Seferbekov, S. (2020). DFDC winning solution (first place, DeepFake Detection Challenge). GitHub. https://github.com/selimsef/dfdc_deepfake_challenge
- [12] Dolhansky, B., Howes, R., Pflaum, B., Baram, N., and Ferrer, C. C. (2019). The DeepFake Detection Challenge (DFDC) Preview dataset. <https://arxiv.org/abs/1910.08854>
- [13] Frank, J., Eisenhofer, T., Schonherr, L., Fischer, A., Kolossa, D., and Holz, T. (2020). Leveraging frequency analysis for deep fake image recognition. ICML. <https://arxiv.org/abs/2003.08685>
- [14] Dolhansky, B., Bitton, J., Pflaum, B., Lu, J., Howes, R., Wang, M., and Ferrer, C. C. (2020). The DeepFake Detection Challenge (DFDC) dataset. <https://arxiv.org/abs/2006.07397>